Please complete all of the following problems.
Show all your work.
Consult the textbook if you need more information or are unfamiliar with the question.

9.1 In general terms, how does each of the following atomic properties influence the metallic character of the main-group elements in a period?
(a) Ionization energy
(b) Atomic radius
(c) Number of outer electrons
(d) Effective nuclear charge

9.4 Which member of each pair is *more* metallic?
(a) Na or Cs (b) Mg or Rb (c) As or N

9.6 State the type of bonding—ionic, covalent, or metallic—you would expect in each: (a) $CsF(s)$; (b) $N_2(g)$; (c) $Na(s)$.

9.8 State the type of bonding—ionic, covalent, or metallic—you would expect in each: (a) $O_3(g)$; (b) $MgCl_2(s)$; (c) $BrO_2(g)$.

9.10 Draw a Lewis electron-dot symbol for each atom:
(a) Rb (b) Si (c) I

9.14 Give the group number and general electron configuration of an element with each electron-dot symbol:
(a) $\cdot\ddot{X}:$ (b) $\dot{X}\cdot$

9.17 In general, how does the lattice energy of an ionic compound depend on the charges and sizes of the ions?

9.20 Use condensed electron configurations and Lewis electron-dot symbols to depict the monatomic ions formed from each of the following atoms, and predict the formula of the compound the ions produce:
(a) Ba and Cl (b) Sr and O (c) Al and F (d) Rb and O

9.22 Identify the main group to which X belongs in each ionic compound formula: (a) XF_2; (b) MgX; (c) X_2SO_4.

9.24 Identify the main group to which X belongs in each ionic compound formula: (a) X_2O_3; (b) XCO_3; (c) Na_2X.

9.26 For each pair, choose the compound with the higher lattice energy, and explain your choice:
(a) BaS or CsCl (b) LiCl or CsCl

9.28 For each pair, choose the compound with the lower lattice energy, and explain your choice:
(a) CaS or BaS (b) NaF or MgO

9.30 Use the following to calculate the $\Delta H^0_{lattice}$ of NaCl:

$Na(s) \longrightarrow Na(g)$	$\Delta H^0 = 109$ kJ
$Cl_2(g) \longrightarrow 2Cl(g)$	$\Delta H^0 = 243$ kJ
$Na(g) \longrightarrow Na^+(g) + e^-$	$\Delta H^0 = 496$ kJ
$Cl(g) + e^- \longrightarrow Cl^-(g)$	$\Delta H^0 = -349$ kJ
$Na(s) + \frac{1}{2}Cl_2(g) \longrightarrow NaCl(s)$	$\Delta H^0_f = -411$ kJ

Compared with the lattice energy of LiF (1050 kJ/mol), is the magnitude of the value for NaCl what you expected? Explain.

9.33 Born-Haber cycles were used to obtain the first reliable values for electron affinity by considering the EA value as the unknown and using a theoretically calculated value for the lattice energy. Use a Born-Haber cycle for KF and the following values to calculate a value for the electron affinity of fluorine:

$K(s) \longrightarrow K(g)$	$\Delta H^0 = 90$ kJ
$K(g) \longrightarrow K^+(g) + e^-$	$\Delta H^0 = 419$ kJ
$F_2(g) \longrightarrow 2F(g)$	$\Delta H^0 = 159$ kJ
$K(s) + \frac{1}{2}F_2(g) \longrightarrow KF(s)$	$\Delta H^0_f = -569$ kJ
$K^+(g) + F^-(g) \longrightarrow KF(s)$	$\Delta H^0 = -821$ kJ

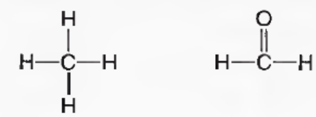
9.35 Define bond energy using the H—Cl bond as an example. When this bond breaks, is energy absorbed or released? Is the accompanying ΔH value positive or negative? How do the magnitude and sign of this ΔH value relate to the value that accompanies H—Cl bond formation?

9.39 Using the periodic table only, arrange the members of each of the following sets in order of increasing bond *strength:*
(a) Br—Br, Cl—Cl, I—I (b) S—H, S—Br, S—Cl
(c) C=N, C—N, C≡N

9.44 The text points out that, for similar types of substances, one with weaker bonds is usually more reactive than one with stronger bonds. Why is this generally true?

9.46 Which of the following gases would you expect to have the greater heat of combustion per mole? Why?

methane or formaldehyde



9.52 Describe the vertical and horizontal trends in electronegativity (EN) among the main-group elements. According to Pauling's scale, what are the two most electronegative elements? The two least electronegative elements?

9.54 Is the H—O bond in water nonpolar covalent, polar covalent, or ionic? Define each term, and explain your choice.

Electronegativity (EN)

9.57 Using the periodic table only, arrange the elements in each set in order of *increasing* EN: (a) S, O, Si; (b) Mg, P, As.

9.59 Using the periodic table only, arrange the elements in each set in order of *decreasing* EN: (a) N, P, Si; (b) Ca, Ga, As.

9.65 Are the bonds in each of the following substances ionic, nonpolar covalent, or polar covalent? Arrange the substances with polar covalent bonds in order of increasing bond polarity:
(a) S_8 (b) RbCl (c) PF_3 (d) SCl_2 (e) F_2 (f) SF_2

9.67 Rank the members of each set of compounds in order of *increasing* ionic character of their bonds. Use *polar arrows* to indicate the bond polarity of each:
(a) HBr, HCl, HI (b) H_2O, CH_4, HF (c) SCl_2, PCl_3, $SiCl_4$

9.70 (a) List four physical characteristics of a solid metal.
(b) List two chemical characteristics that lead to an element being classified as a metal.